[14]:
```
"""
13. Employee Bonus Calculator
Accept years of experience and performance rating. If experience is at least 5 years, check the rating.
Give a 20% bonus for an excellent rating and 10% for a good rating.
If experience is below 5 years, give a 5% bonus only for an excellent rating.
"""


experience = int(input("Enter years of experience: "))
rating = input("Enter performance rating: ")

if experience >= 5:
    if rating == "excellent":
        print("Bonus: 20%")
    elif rating == "good":
        print("Bonus: 10%")
    else:
        print("No Bonus")
else:
    if rating == "excellent":
        print("Bonus: 5%")
    else:
        print("No Bonus")
```

```
Enter years of experience:  5
Enter performance rating:  excellent
Bonus: 20%
```

[1]:
```
'''14. Password Strength Checker
Accept a password. If its length is at least 8 characters, check whether it contains a digit.
If it does, check whether it contains an uppercase letter. Classify it as strong or moderate based on these conditions.
Otherwise, classify it as weak.'''

pw = input("Enter a password")
uc=0
lc=0
sp=0
nd=0
if(len(pw)>=8):
    for i in pw:
        if i.isupper():
            uc+=1
        elif i.islower():
            lc+=1
        elif i.isdigit():
            nd+=1
        else:
            sp+=1

else:
    print("should be > 7 characters")
if(uc>=1 and lc>=1 and nd>=1 and sp>=1):
    print("strong password")
else:
    print("weak password")
```

```
Enter a password hema
should be > 7 characters
weak password
```

[3]:
```
'''15. Hospital Consultation Priority
Accept a patient's temperature and oxygen saturation. If oxygen saturation is below 90%, classify as high priority.
Otherwise, check whether the temperature is at least 39°C.
If so, classify as urgent; otherwise, classify as normal priority. (Simplified programming exercise, not a medical triage tool.)'''

temp = float(input("enter a temperature"))
o = float(input("Enter oxygen saturation"))
if o <90:
    print("high priority")
else:
    if temp >= 39 :
        print("urgent")
    else:
        print("normal priority")
```

```
enter a temperature 89
Enter oxygen saturation 39
high priority
```

[4]:
```
'''16. Income Tax Slab Calculator
Accept annual taxable income. If income is up to ₹3 lakh, tax is 0%. Otherwise,
check whether it is up to ₹7 lakh and apply a 5% rate to income above ₹3 lakh.
For income above ₹7 lakh, apply progressive slabs of 0%, 5%, and 10% to the respective portions. Calculate the total tax.'''

i = int(input("Enter annula taxable income"))
a=0;
if(i <= 300000):
    tax=0
elif i <= 700000:
    tax = (i-300000)*5/100
else:
    tax1=0
    tax2 = 400000*5/100
    tax3 = (i-700000)*10/100
    tax = tax1+tax2+tax3
print("total income tax",tax)
```

```
Enter annula taxable income 400000
total income tax 5000.0
```

[6]:
```
'''17. College Admission
Accept marks in Mathematics, Physics, and Chemistry. If Mathematics is at least 60, check Physics.
If Physics is at least 50, check Chemistry.
If Chemistry is at least 50 and the total is at least 180, display "Eligible"; otherwise, display "Not Eligible".'''

p = int(input("Enter physics marks"))
m = int(input("Enter maths marks"))
c = int(input("Enter chemistry marks"))
total=p+m+c
if(m >=60):
    if(p>=50):
        if(c>=50 and total >=180):
            print("eligible")
        else:
            print("Not Eligible")
    else:
        print("Not Eligible")
else:
    print("Not Eligible")
```

```
Enter physics marks 90
Enter maths marks 90
Enter chemistry marks 90
eligible
```

[8]:
```
"""18. Online Food Delivery
Accept order amount, distance, and whether the customer has a membership. If the order amount is at least ₹500,
check whether the distance is within 5 km. Apply free delivery if within 5 km;
otherwise, charge ₹40. If the order is below ₹500, check membership status and calculate delivery charges accordingly."""


amt = int(input("Enter order amount"))
dis = float(input("Enter distance"))
membership = input("do you have a membersihp(yes/no): ")
if amt >= 500:
    if dis <= 5:
        delivery = 0
    else:
        delivery = 40
else:
    if(membership=="yes"):
        delivery = 20
    else:
        delivery = 50
total = amt+delivery
print("delivery charge : ",delivery)
print("total ampunt : ",total)
```

```
Enter order amount 400
Enter distance 3
do you have a membersihp(yes/no):  yes
delivery charge :  20
total ampunt :  420
```

[ ]:
```
Accept age, smoking status, and annual income.
If the applicant is under 30, check smoking status to determine the risk category.
If age is 30 or above, check income and smoking status to classify the applicant as low, medium, or high risk.
Define your own clear rules for each category.

Age below 30:
- Non-smoker → Low Risk
- Smoker → Medium Risk
Age 30 or above:
- Income ₹5,00,000 or more and non-smoker → Low Risk
- Income ₹5,00,000 or more and smoker → Medium Risk
- Income below ₹5,00,000 and non-smoker → Medium Risk
- Income below ₹5,00,000 and smoker → High Risk
"""

age = int(input("Enter age: "))
smoking = input("Are you a smoker? (yes/no): ")
income = float(input("Enter annual income: "))

if age < 30:
    if smoking == "yes":
        print("Medium Risk")
    else:
        print("Low Risk")
else:
    if income >= 500000:
        if smoking == "yes":
            print("Medium Risk")
        else:
            print("Low Risk")
    else:
        if smoking == "yes":
            print("High Risk")
        else:
            print("Medium Risk")
```

```
Enter age:  30
Are you a smoker? (yes/no):  no
Enter annual income:  300000
Medium Risk
```

[10]:

```
"""20. Mini Banking System
Accept account type, account balance, and transaction type (deposit or withdrawal).
If the account type is savings, check the transaction type.
For withdrawals, check minimum balance requirements; for deposits, validate the deposit amount.
If the account type is current, apply a different minimum balance rule.
"""

acc_type = input("account type(savings/current): ")
balance = float(input("Enter account balance "))
t_type= input("enter transaction type (deposit/withdraw")
amount= float(input("Enter transaction amount"))
if acc_type == "savings":
    if t_type == "deposit":
        if amount > 0:
            balance+=amount
            print("deposit successfully")
        else:
            print("invalid deposit amount")
    elif t_type=="withdraw":
        if(balance - amount >= 1000):
            balance-=amount
            print("withdraw successfully")
        else:
            print("minimum balance required")
    else:
        print("invalid rtansaction type")
elif account == "current":
    if transaction == "deposit":
        if amount > 0:
            balance = balance + amount
            print("Deposit Successful")
        else:
            print("Invalid Deposit Amount")

    elif transaction == "withdrawal":
        if balance - amount >= 5000:
            balance = balance - amount
            print("Withdrawal Successful")
        else:
            print("Minimum Balance Required")

    else:
        print("Invalid Transaction Type")

else:
    print("Invalid Account Type")

print("Available Balance:", balance)
```

```
account type(savings/current):  savings
Enter account balance  30000
enter transaction type (deposit/withdraw withdraw
Enter transaction amount 10000
withdraw successfully
Available Balance: 20000.0
```